System(3X±1)/4 Framework for the Collatz (3x+1) Problem : A Definitive Proof of the Collatz Conjecture

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Abstract

We define two systems, the System(3X±1)/4 and the CTUHSK(3x+1)/4 system; similar to the Collatz (3x+1) system, with modified operations. We establish the convergence of the System(3X±1)/4 to its trivial-cycle {(1⇐(2)⇐4)}; using the persistent local characteristic property of odd-to-odd monotonicity in its defining relation. Therefore, the transitive-closure of the subset, say {1,5,3}; under the inverse mapping of System(3X±1)/4; represented by the nodes of the system arborescence rooted at its trivial-cycle {(1⇐(2)⇐4)}; is the entire set of positive integers. We present a transformation; from System(3X±1)/4 to the CTUHSK(3x+1)/4 system; based on a modular arithmetic framework for the pruning-&-joining atomic dual-operation on infinite binary arborescence. We show that every such pruning-&-joining atomic dual-operation preserves the infinite binary(inward)arborescence structure of the system. The transformation process

itself is shown to be a Cauchy sequence of infinite binary(inward)arborescence under the Cantor metric space; that achieves a sequential monotonic decrease, by iterative elimination, of the symmetric difference between these two systems; guaranteeing the convergence of the transformation process. Therefore, the transitive-closure of the subset, say {1,5,3} under the inverse mapping of CTUHSK(3x+1)/4 system is exactly the same as the transitive-closure of that subset under the inverse mapping of System(3X±1)/4; which has been shown to be the entire set of all natural numbers. This establishes the convergence of the CTUHSK(3x+1)/4 system; and therefore, the convergence of the Collatz (3x+1) sequence; thus, providing a definitive proof of the Collatz Conjecture.

This paper also presents Halemane-Conjecture on the Collatz sequence, that the upper bound on the number of odd (3x+1) operations required to reach its trivial-cycle {(1⇐2⇐4)} starting from any given positive integer and moving along the Collatz sequence, is the given number itself; the triad {(31⇐41⇐27)} being an exceptional limiting case.

Keywords: System(3X±1)/4; CTUHSK(3x+1)/4 system;
Halemane's Conjecture on the Collatz (3x+1) system;
Collatz-Thwaites-Ulam-Hasse-Syracuse-Kakutani (CTUHSK) Sequence;
Collatz-Thwaites-Ulam-Hasse-Syracuse-Kakutani (CTUHSK) Conjecture;
infinite binary(inward)arborescence;
pruning-&-joining atomic dual-operation;

AMS MSC Mathematics Subject Classification: 11B83; 40A99; 05C20; 05C63;

1. Introduction

The Collatz-Thwaites-Ulam-Hasse-Syracuse-Kakutani (CTUHSK) Conjecture, simply referred to as the *Collatz Conjecture* [§1] states that the Collatz (3x+1) Sequence, also referred to as the Collatz Sequence, converges to its trivial-cycle; that is, {(1⇐2⇐4)}, starting from any positive integer.

Lagarias [§2]&[§3] gives an exhaustive annotated bibliography, whereas Lagarias [§4]&[§5] gives an elaborate overview of the Collatz Problem, also referred to as the "3x+1 problem", referring to it as "The Ultimate Challenge". Guy [§6] has been somewhat pessimistic in advising researchers "Don't Try to Solve These Problems". Lagarius [§4]&[§5] refers to the statement by the most revered Mathematician and Number Theory Expert Paul Erdos, who said that - "Mathematics is not yet ready for such problems" - about the Collatz Conjecture. The Collatz (3x+1) Sequence is indeed a highly complex chaotic system, although defined or generated with one of the simplest deterministic well-defined equations, that too, just two in number.

We will not get into any more of the reported literature except the problem definition; but will simply focus on a rather novel approach toward a possible resolution of this amazingly notorious long-standing unresolved problem.

The paper presents a modular arithmetic framework to design a Cauchy sequence of infinite arborescence [§7] under the Cantor metric space; that achieves a sequential monotonically strictly decreasing iterative elimination, of the symmetric difference between the two systems; that is, System(3X±1)/4 and the

CTUHSK(3x+1)/4 system; therefore, establishing that the convergence of the first system implies the convergence of the second system. The CTUHSK(3x+1)/4 system is indeed a redefined rewritten form of the Collatz (3x+1) system; and therefore, this paper provides a definitive proof of the Collatz Conjecture.

2. System(3X±1)/4

System(3X±1)/4 is defined by its immediate-successor relation S(x) as follows:

For any positive integer x;
if $x \equiv 0 (\mathrm{mod} 4)$ then, $S(x) \coloneqq (x/4)$;
elseIf $x \equiv 2 (\mathrm{mod} 4)$ then, $S(x) \coloneqq (x/2)$;
elseIf $\{x \equiv 1 (\mathrm{mod} 4)\}\}$ then, $S(x) \coloneqq (3x+1)$;
elseIf $\{x \equiv 3 (\mathrm{mod} 4)\}\}$ then, $S(x) \coloneqq (3x-1)$; [Eqn.1]

Note that as per the above definition, for every positive integer x; $y \coloneqq S(x)$ is its single unique immediate-successor of x.

In order to gain better understanding of the structure of System(3X±1)/4; we first construct the inverse relation, derived from the above definition; that defines the immediate-predecessor {P(y)} as -

For any given positive integer y;
if $y \in \{[1]_4; [3]_4\}$; then, $\{P(y)\} \coloneqq \{(2y); (4y)\}$;
if $y \in \{[0]_{12}; [2]_{12}; [6]_{12}; [10]_{12}\}$; then, $\{P(y)\} \coloneqq \{(4y)\}$;
if $y \in \{[4]_{12}\}$; then, $\{P(y)\} \coloneqq \{(4y); (y-1)/3\}$;

if $y \in \{[8]_{12}\}$; then, $\{P(y)\} \coloneqq \{(4y); (y+1)/3\}$; [Eqn.2]

Here, $[c]_b$ represents the modular (modulo-b) residue class c that a given positive integer belongs to.

3. Observations on the Structure of System(3X±1)/4

Based on the above defining relations, we observe that the System(3X±1)/4 has the following characteristic properties:

(0) every positive integer x has exactly one immediate-successor y.

(1) every positive odd integer $y \in \{[1]_4; [3]_4\}$ has two immediate-even-predecessors, and no immediate-odd-predecessors.

(2) every positive even integer $y \in \{[0]_{12}; [2]_{12}; [6]_{12}; [10]_{12}\}$ has exactly one *immediate-even-predecessor*, and no immediate-odd-predecessors; such nodes are designated as inactive-nodes.

(3) every positive even integer $y \in \{[4]_{12}; [8]_{12}\}$ has one immediate-even-predecessor and one immediate-odd-predecessor; such nodes are designated as active-nodes.

(4) every positive odd integer $y \in [3]_6 \coloneqq \{[3]_{12}; [9]_{12}\}$; y being an odd-multiple-of-3; is an odd-leaf-node; none of its even predecessors has any immediate-odd-predecessor.

These properties are characteristic of an infinite binary(inward)arborescence structure H for the System(3X±1)/4; with its trivial-cycle {(1⇐(2)⇐4)} as a root. Here, every positive even integer having an immediate-odd-predecessor is called an active-node $y \in \{[4]_{12}; [8]_{12}\}$. Every positive even integer having no immediate-odd-predecessor is called an inactive-node $y \in \{[0]_{12}; [2]_{12}; [6]_{12}; [10]_{12}\}$. Also, every

positive odd multiple of 3 is an odd-leaf-node y $\in \{[3]_{12}; [9]_{12}\}$; that is, none of its even predecessors y $\in \{[0]_{12}; [6]_{12}\}$ can have any immediate-odd-predecessor.

The transitive-closure of the subset, say {1,5,3}; under the inverse mapping P of System(3X±1)/4; is represented by the nodes of this system arborescence; that is.

$$[\{1,5,3\}P+] = \{\text{nodes}(H)\} \qquad [\text{Eqn.3}]$$

4. Convergence of System(3X±1)/4

Note that every application of the (3X±1) operation produces an immediate resultant positive even numbers x$\in\{[0]_4\}$; and therefore every (3X±1) operation *must necessarily be followed* by *at least one* application of the (x/4) operation; that may even be followed by further repetitions of it; leading to an *immediate decrease* in the numerical value of the operand. Therefore, the resulting odd-to-odd subsequence is a *monotonically strictly decreasing odd-to-odd subsequence*; a characteristic property that we refer to as, a *persistent local odd-to-odd monotonicity*. Note that every even-to-odd subsequence is monotonically strictly decreasing by definition. Hence, this immediately establishes our assertion, that –

Assertion-H-1:
The System(3X±1)/4 converges to its trivial-cycle {(1⇐(2)⇐4)}; starting with any positive integer x > 1.

This convergence property, Assertion-H-1; along with the defining relation for System(3X±1)/4; immediately implies our Assertion-H-2; that is –

Assertion-H-2:

The transitive-closure of the subset {1,5,3}; that is, [{1,5,3}P+]; under the inverse mapping P of System(3X±1)/4; represented by the nodes {nodes(H)} of the system arborescence H; is exactly equal to the entire set of positive integers N; that is,

$$[\{1,5,3\}P+] = \{\text{nodes}(H)\} = N. \qquad \text{[Eqn.4]}$$

5. Collatz (3x+1) system re-written as CTUHSK(3x+1)/4 system

The Collatz(3x+1)/2System can be defined by its immediate-successor relation f(x) for the (classical/canonical) Collatz sequence that can be written as –

For any positive integer x;
if $x \equiv 0 (\text{mod} 2)$ then, $f(x) := (x/2)$;
elseIf $\{(x \equiv 1 (\text{mod} 2)\}$ then, $f(x) := (3x+1)$; [Eqn.5]

Note that as per [Eqn.5] above; for every positive integer x there exists a single unique immediate-successor $y := f(x)$.

Also note that the immediate-successor of any positive odd number x of the form (4m+1) in the Collatz sequence is a positive even number of the form (12m+4); that is, a number $y \in \{[0]_4\}$ that is always divisible by 4 *at least once*, and therefore the combined operation can be denoted by a single ((3x+1)/4) operation; which leads to a clear decrease in the numerical value for the resultant odd number.

6. Critical-Issue, Critical-(Decision)-Node, Detour-Link, Detour-Node

Observe that the immediate-successor of any positive odd number x of the form (4m-1) in the Collatz sequence is a positive even number y of the form (12m-2); that is, a number y $\in \{[2]_4\}$ that is always divisible by 2 *exactly once* only. The combined operation can be denoted by a single ((3x+1)/2) operation, with its immediate resultant being a positive odd number z of the form (6m-1); with an increase in the numerical value of the operand. We designate this specific step {(12m-2)⇐(4m-1)} in the Collatz system as a critical *detour-link*. The critical detour-link is the very source for the possible ambiguity (*critical-issue*) regarding the convergence of the Collatz sequence. We designate the initial-node (4m-1) of a detour-link as its *critical-*(*decision*)*-node*; and its terminal-node (12m-2) as its *detour-node*. This *critical-issue* is addressed appropriately here in this paper.

As a first step, the Collatz (3x+1) system originally defined by [Eqn.3] can now be re-written as an equivalent CTUHSK(3x+1)/4 system; with a special single application of a (x/2) operation in such specific cases; whenever the operand is a positive even number of the form (12m-2) arising as the immediate resultant of the application of the (3x+1) operation on a positive odd number of the form (4m-1).

The Collatz (3x+1) system; now re-written & renamed as its equivalent CTUHSK(3x+1)/4 system; is defined by its redefined immediate-successor relation:

For any positive integer x;
if $x \equiv 0 (\mathrm{mod} 4)$ then, $F(x) := (x/4)$;
elseIf $x \equiv 2 (\mathrm{mod} 4)$ then, $F(x) := (x/2)$;

elseIf {(x≡1(mod4)) ∨ (x≡3(mod4))} then, F(x)≔ (3x+1); [Eqn.6]

Assertion-C-1:

The CTUHSK(3x+1)/4 system; by its very definition as in [Eqn.6] above; is exactly identical to the Collatz (3x+1) system; noting an insignificant difference, that a division by 4 skips (double-jump) over the corresponding intermediate positive even number that would be reached and passed through explicitly, by a division by 2.

Note that as per the above definition of CTUHSK(3x+1)/4 system; every positive integer x; has its single unique immediate-successor, y ≔ F(x).

In order to gain better understanding of the structure of CTUHSK(3x+1)/4 system; we construct the inverse relation, defined by the immediate-predecessor relation {R(y)}; derived from the above definition; that is -

For any positive integer y;
if y ∈{$[1]_4$; $[3]_4$}; then, {R(y)} ≔ {(2y); (4y)};
if y ∈{$[0]_{12}$; $[2]_{12}$; $[6]_{12}$; $[8]_{12}$}; then, {R(y)} ≔ {(4y)};
if y ∈{$[4]_{12}$; $[10]_{12}$}; then, {R(y)} ≔ {(4y); (y-1)/3}; [Eqn.7]

7. Observations on the Structure of CTUHSK(3x+1)/4 system

Based on the above defining relations we observe that the CTUHSK(3x+1)/4 system has the following characteristic properties:
(0) every positive integer x has exactly one immediate-successor y.

(1) every positive odd integer y ∈ {$[1]_4$; $[3]_4$} has two immediate-even-predecessors, and no immediate-odd-predecessors.

(2) every positive even integer y ∈ {$[0]_{12}$; $[2]_{12}$; $[6]_{12}$; $[8]_{12}$} has exactly one *immediate-even-predecessor*, and no immediate-odd-predecessors; such nodes are designated as inactive-nodes.

(3) every positive even integer y∈ {$[4]_{12}$; $[10]_{12}$} has one immediate-even-predecessor and one immediate-odd-predecessor; such nodes are designated as active-nodes.

(4) every positive odd integer y ∈ $[3]_6$ ≔ {$[3]_{12}$; $[9]_{12}$}; y being an odd-multiple-of-3; is an 'odd-leaf-node'; none of its even predecessors has any immediate-odd-predecessor.

These properties are characteristic of an infinite binary(inward)arborescence structure C for the CTUHSK(3x+1)/4 system; with its trivial-cycle {(1⇐2⇐4)} as a root. Here, every positive even integer having an immediate-odd-predecessor is called as an active-node y ∈ {$[4]_{12}$; $[10]_{12}$}; whereas every positive even integer having no immediate-odd-predecessor is called an inactive-node y ∈ {$[0]_{12}$; $[2]_{12}$; $[6]_{12}$; $[8]_{12}$}. Also, every positive odd multiple of 3 is an odd-leaf-node y ∈ {$[3]_{12}$; $[9]_{12}$}; none of its even predecessors y ∈ {$[0]_{12}$; $[6]_{12}$} can have any immediate-odd-predecessor.

From these observations, we derive Assertion-C-1 as follows -

Assertion-C-1:

The transitive-closure of the subset {1,5,3}; that is, [{1,5,3}R+]; under the inverse mapping R of CTUHSK(3x+1)/4 system; represented by the nodes {nodes(C)} of the system arborescence C; that is,

$$[\{1,5,3\}R+] = \{\text{nodes}(C)\} \qquad [\text{Eqn.8}]$$

However, we need to establish that $[\{1,5,3\}R+]$ is exactly the same as $[\{1,5,3\}P+]$.

8. Observations on the Critical Difference between H and C

The difference between the System(3X±1)/4 and the CTUHSK(3x+1)/4 system; as also between the system arborescence H and the system arborescence C; as observed from their defining relations, corresponds to the node (4m-1) and its connecting links. We designate such the node (4m-1) as the critical-decision-node. The critical-(decision)-node (4m-1) $\in \{[3]_{12}; [7]_{12}; [11]_{12}\}$ is linked to the corresponding node (12m-4) $\in \{[8]_{12}\}$ through the link $\{(12m\text{-}4) \Leftarrow (4m\text{-}1)\} \in H$ in System(3X±1)/4; whereas it is linked to the node (12m-2) $\in \{[10]_{12}\}$ through the link $\{(12m\text{-}2) \Leftarrow (4m\text{-}1)\} \in C$ in the CTUHSK(3x+1)/4 system. We designate the node (12m-4) as the shortcut-node and that link $\{(12m\text{-}4) \Leftarrow (4m\text{-}1)\} \in H$ as the shortcut-link. We designate that node (12m-2) as the detour-node and that link $\{(12m\text{-}2) \Leftarrow (4m\text{-}1)\} \in C$ as the detour-link. Therefore, the symmetric difference between system arborescence H and system arborescence C is given by the set of all pairs of links of the form {shortcut-link; detour-link}; of which the component at the level associated with the label say, m corresponding to the critical-decision-node (4m-1) is, specifically, the pair of links:

$$\{(12m\text{-}4) \Leftarrow (4m\text{-}1) \in H\} \cup \{(12m\text{-}2) \Leftarrow (4m\text{-}1) \in C\}. \qquad [\text{Eqn.9}]$$

9. From System Arborescence H to System Arborescence C

Perform the following transformation, that is, an iterative process based on the following well-defined pruning-&-joining atomic dual-operation; on the infinite binary(inward)arborescence H of the System(3X±1)/4.

Transformation Process H2C:

Initialization step 0: $H_0 = H$.

Iterative step k: $H_{k+1} = [H_k \setminus \{(12k\text{-}4)\Leftarrow(4k\text{-}1)\}] \cup \{(12k\text{-}2)\Leftarrow(4k\text{-}1)\}$;

The following three observations regarding this iterative process, together leads to the Assertion-C-1 as stated below.

Observation-H2C-1:

Every pruning-&-joining atomic dual-operation preserves the set of nodes; neither any nodes are added nor any nodes are removed.

Observation-H2C-2:

The iteration k removes the shortcut-link $\{(12k\text{-}4)\Leftarrow(4k\text{-}1)\} \in H_k$; and adds the detour-link $\{(12k\text{-}2)\Leftarrow(4k\text{-}1)\} \in C$. This operation results in the preserving the out-degree of the critical-decision-node (4k-1); a decrease in the in-degree of the shortcut-node from two to one; an increase in the in-degree of the detour-node from zero to one; the in-degree & out-degree of all the other nodes remain unaffected. Therefore, the infinite binary(inward)arborescence structure remains unaffected.

Observation-H2C-3:

As we move forward along this iterative sequence; each & every specific node-triad {critical-decision-node, shortcut-node, detour-node} as well as the corresponding link-pair {shortcut-link, detour-link} get *processed exactly once* and then remain frozen/untouched in the entire transformation process. This satisfies the Cauchy sequence requirement on Cantor metric space (associated with the two infinite binary(inward)arborescence, H_k and C); that is, resulting in a sequential monotonic reduction by iterative elimination of the symmetric difference between H_k and C; therefore, achieving a monotonically strictly reducing Cantor metric at every iteration. This establishes the convergence of the infinite series H_k, k=1,2,3, etc. to C; and in turn the following assertion -

Assertion-C-1:

The transitive-closure of the subset, say {1,5,3}; that is, [{1,5,3}R+]; under the inverse mapping R of the resultant system CTUHSK(3x+1)/4; represented by the nodes {nodes(C)} of the resultant system arborescence C; is the exactly same as the transitive-closure of the same subset {1,5,3}; under the inverse mapping P of the original system original System(3X±1)/4; represented by the nodes {nodes(H)} of the original system arborescence H; that is,

$$\{\text{nodes(C)}\}=[\{1,5,3\}R+] = [\{1,5,3\}P+]=\{\text{nodes(H)}\}. \qquad \text{[Eqn.10]}$$

However, Assertion-H-2 above, already established that the transitive-closure of the subset {1,5,3}; that is, [{1,5,3}P+]; under the inverse mapping P of

System(3X±1)/4; represented by the nodes {nodes(H)} of the original system arborescence H; is the entire set of positive integers N. that is,

$$\{\text{nodes(C)}\}=[\{1,5,3\}R+] = [\{1,5,3\}P+]=\{\text{nodes(H)}\} = N. \qquad \text{[Eqn.11]}$$

10. From System Arborescence C to System Arborescence H

A transformation process from system arborescence C to system arborescence H can be designed exactly similar to that described above (from H to C); with the same node-triad {critical-decision-node, shortcut-node, detour-node} as well as the corresponding link-pair {detour-link, shortcut-link}; this time, replacing every detour-link for the corresponding shortcut-link; with exactly similar convergence properties. It is left out simply because it is not specifically necessary/required for the purpose of establishing the desired results.

11. Summary

Two systems have been designed, System(3X±1)/4 and the CTUHSK(3x+1)/4; for the study of the pruning-&-joining operations on infinite arborescence using a modular arithmetic framework. The convergence of System(3X±1)/4 is established using the persistent local odd-to-odd monotonicity property of its defining relation. A transformation (reduction) from System(3X±1)/4 to CTUHSK(3x+1)/4 system; has been designed, based on an iterative process, using a modular arithmetic framework to define the pruning-&-joining atomic dual-operation performed on the infinite binary(inward)arborescence structure of the system. Each pruning-&-

joining operation preserves the system arborescence structure. It is shown that the transformation process is a Cauchy sequence of infinite arborescence under the Cantor metric space; that achieves a sequential minimization of the symmetric difference between these two systems; therefore, guaranteeing the convergence. Hence, the transitive-closure of the subset, say {1,5,3} under the inverse mapping of CTUHSK(3x+1)/4 system is exactly the same as the transitive-closure of that subset under the inverse mapping of System(3X±1)/4; which has been shown to be the entire set of all natural numbers. This establishes the convergence of the CTUHSK(3x+1)/4 system. This result immediately implies the convergence of the Collatz sequence and therefore provides a definitive proof of the Collatz Conjecture.

The Halemane-Conjecture on the Collatz (3x+1) sequence states that the maximum limit on number of odd (3x+1) operations required to reach its trivial-cycle {(1⇐2⇐4)}; starting from any given positive integer and moving along the Collatz sequence; is the given number itself; with the triad {(31⇐41⇐27)} as an exceptional limiting case.

12. Recommended Reading

[§1]. Wikipedia Page – https://en.wikipedia.org/wiki/Collatz_conjecture

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13. Acknowledgment

The author wishes to acknowledge with gratitude the excellent openminded understanding & appreciation of the work by the Editorial Board & the Reviewers Team, following the most basic principle of paying credit wherever-&-whenever it is rightfully due, without any misplacement. Note that the Collatz Conjecture has been referred to as the Collatz-Thwaites-Ulam-Hasse-Syracuse-Kakutani

(CTUHSK) Conjecture and the CTUHSK(3x+1)/4 system has been named so to acknowledge all the known original sources thereof. Similarly, this author wishes that the System(3X±1)/4 be henceforth referred to as Halemane(3X±1)/4System.

14. A Confession and An Assertion

I must necessarily confess that the core idea behind this work is so stunningly and elusively simple, that one may simply be taken aback in a profound wonder-struck jaw-drop-silence, possibly with an after-thought:
"OMG, how could it be that it never flashed on me any time earlier"!
However, I assert that क्लिष्टविशयसरलीकरणमेव पाण्डित्यलक्षणम् | न तु सरलविषयसंक्लिष्टीकरणम् || That is, *to make the complex simple is a mark of mastery*; *not to make the simple complex*.

15. Declaration Regarding Affiliation and Funding

I, Dr(Prof) Keshava Prasad Halemane, hereby declare that I am a Professor retired as on 2017JAN31 from National Institute of Technology Karnataka Surathkal India, and I am not affiliated to any institution or organization or corporation or any other agency or whatever. This research work has been conducted entirely by me on my own as an Independent Researcher, and that I have not received any funding from any source other than my own savings, and I do not have any obligations or encumbrances of any kind, neither financial nor legal nor of any other kind, regarding the contents of the manuscript - of which I am the original author and creator. Also, I hereby declare there is no conflict of interests.

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17. Dedication

To my **ಅಜ್ಜ**(ajja) Karinja Halemane Keshava Bhat & **ಅಜ್ಜಿ**(ajji) Thirumaleshwari, **ಅಪ್ಪ**(appa) Shama Bhat & **ಅಮ್ಮ**(amma) Thirumaleshwari, for their *teachings through love*, that *quality matters more than quantity*; to my wife Vijayalakshmi for her *ever consistent love & support*; to my daughter Sriwidya.Bharati and my twin sons Sriwidya.Ramana & Sriwidya.Prawina for their *love & affection.*